

Muhammad Adeel Arif

Senior Engineer (Power Systems), Dominion Energy VA

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PROFESSIONAL SUMMARY

Senior Electrical Engineer with 10+ years of experience across utility, industrial, and research environments, specializing in grid resiliency, distributed energy systems, and intelligent grid automation. Expertise in ADMS grid automation (FLISR, UFLS, VVO, EVR), distributed-generation modeling, and mission-critical power design integrating batteries, inverters, and microgrid assets.

EXPERTISE

Smart Grid Design | Grid Automation for DER dominated grids | ADMS/DERMS | FLISR | Protection & Control | Energy Storage Systems | GFM/GFL Inverter Control | System Modeling (PSS/E, PowerWorld, OpenDSS, GridLAB-D, HELICS, OPAL-RT, MATLAB) | SCADA Integration | Optimization | SAIDI/SAIFI | ML/RL | HIL/EMT

EXPERIENCE

Senior Engineer

Dominion Energy, VA

Nov 2025 – Present

Richmond, VA USA

- Implementing and validating ORACLE's Mixed-Integer Programming (MIP) based FLISR (Fault location, Isolation and service Restoration) program for self healing grids and 3-phase power-flow models on 34.5 kV feeders; improved restoration time by ~18% and enhanced compliance with IEEE 1366 SAIDI/SAIFI reliability metrics.
- Technical Engineering Support for integration of UFLS (Under frequency load shedding), EVR (emergency Voltage Reduction), VVO (Volt Var Optimization), and PCLLRW programs—cross-protocol optimization improving feeder reliability for 100,000+ customers for PJM compliance.

Engineer

Dominion Energy, VA

April 2025 – Oct 2025

Richmond, VA USA

- Configured and tested ADMS components based on IEC 61970/61968 (CIM); ensured interoperability with real-time SCADA and outage-management systems. Work parallels mission-critical data-center energy-management architectures, focusing on redundancy, resiliency, and rapid restoration.

Research Assistant

Center for Resilient Energy and Autonomous Technologies in Engineering (CREATE), UVM

Aug 2023 – April 2025

Burlington, VT, USA

- Modeled and tested dynamics for Unified (GFM/GFL) DER inverters, performed data-driven stability assessment under Fault Ride-Through Modes for grid codes GC0137, FERC 827, 828, and 842. Validation results on OPAL-RT Real-time Simulator for Hybrid Energy System (HES) project, funded by the Department of Energy (DOE).
- Led machine-learning/Data-Driven Modeling of Lithium-Ion Battery Lifecycle: Predicting Capacity Degradation under Fast-Charging Conditions Using Machine Learning and Deep Learning (RF, SVM, LSTM, BiLSTM). Authored technical documentation and publications supporting energy-system flexibility and sustainable-grid research.

Lecturer - Electrical Power Systems

NED University of Engineering and Technology

Mar 2021 – Aug 2023

Karachi, Pakistan

- Taught and mentored 1900 + students in Control Systems, Power Electronics, and Programming (C++).
- Guided senior-year research projects integrating renewable generation and control algorithms for distributed grids.

Research Assistant

Energy Research Center, Arizona State University

Jan 2019 – May 2019

Tempe, AZ, USA

- Designed and modeled intelligent islanding detection schemes for Grid-following inverter-based systems at the Machine Learning Laboratory for Power Systems, Arizona State University, resulting in an IEEE-Access transaction.

Research Assistant

Advanced Smart Grid Computation Lab, U.S Pakistan Center for Advanced Studies, NUST

Aug 2017 – May 2021

Pakistan

- Developed a state-of-the-art data-driven Autoencoder-based online active islanding detection system for inverter-based distributed generation using unsupervised anomaly detection, achieving an F1-score of 0.991, dependability index of 0.998, security index of 0.995, complying with IEC 61000-3-2 and IEEE Standard 1547.

Executive Engineer

Nov 2015 – Aug 2017

FFBL Power Company Ltd.

Karachi, Pakistan

- Actively contributed to the execution of a 118 MW Coal Power Project, including commissioning and design review of four steam turbine generator units, with a total project value of 60.04 million USD.
- Supervised the installation and testing of generator protection and control systems (GE G60), generator excitation control systems (BASLER DECS250), and MV-LV substation switchgear and transformers (Schneider, ABB). Actively participated in primary design engineering tasks, including RFQ evaluation, design of interconnections and interlocking schemes.

Engineer

Feb 2015 – Nov 2015

Schneider Electric Ltd.

Karachi, Pakistan

- Project management and planning for MV-LV switchgears, including document control, design review, and project tracking through tendering, procurement, design engineering, production, and factory acceptance tests.
- Managed projects for Aisha Steel Mills, K-Electric, Engro Fertilizers, and FFBL Power Company Ltd.

Graduate Student Senator - Graduate Student Senate, University of Vermont

2023 - 2025

ASU Energy Research Exchange Research Scholar - USAID

2019 - 2019

General Secretary – IEEE PES USPCAS-E

2017 - 2018

EDUCATION**University of Vermont**

VT, USA

Master's in Electrical Engineering (Energy Systems Optimization), GPA: 3.51/4.00

Dec 2024

U.S.-Pakistan Center for Advanced Studies in Energy, NUST

Pakistan

Master's in Electrical Engineering (Power System Protection), GPA: 3.50/4.00

2021

Arizona State University

AZ, USA

Research Exchange (Machine Learning for Power Systems), Grade: A

2019

NED University of Engineering and Technology

Pakistan

Bachelor of Electrical Engineering, GPA: 3.13/4.00

2015

PUBLICATIONS

- J1:** [A. Arif, K. Imran, Y. Weng, and Q. Cui, "Islanding Detection for Inverter-Based Distributed Generation Using Unsupervised Anomaly Detection," in IEEE-Access, 2021.](#)
- C1:** [S. Shaikh, A. Arif, M. Aman "Estimation of Technical Losses on Transmission Systems Using a Neural Network Prognosis Algorithm \(NNPA\)", 2023.](#)
- C2:** [W. Jalil, W. Ishtiaque, A. Arif "A Review of Islanding Detection Techniques for Inverter-Based Distributed Generation", 2023.](#)
- C3:** [A. Arif, S. Sheikh, Danyal Arif, Safwan Wshah "Data-Driven Capacity Degradation Modeling for Commercial Lithium-Iron Phosphatase Batteries under Fast-charging Conditions Using LSTM", in International Conference on Emerging Power Technologies \(ICEPT\) 2025.](#)
- C4:** [U. Ahmed, K. Imran, A. Arif, F. Mumtaz "Field Programmable Gate Array based multi-algorithm relay for anti-islanding protection in smart grids", in 6th International Conference on Energy, Environment and Sustainable Development 2022 \(EESD 2022\)](#)
- P1:** [A. Arif "Site testing protocols for a dry-type transformer under IEEE C57.12", 2016](#)

TRAININGS AND CERTIFICATIONS

PNNL - CREATE	Energy Hackathon: Hands-on training for OpenDSS, GridLab-D, and HELICS
USAID - ASU	Energy Research Training at Arizona State University.
SEECs-NUST	Bootcamp on Artificial Intelligence and Machine Learning.
NCAI-NUST	Deep learning for object detection and recognition.
City & Guilds	Industrial Automation 1 and 2 (PLC, HMI, DSC and SCADA)
SDC-Certificate	Project Management using Primavera Enterprise 6
SDC-Certificate	Certificate in Electrical Design for Hi-Rise Building
World Commonwealth	World Open-Source Leader 2020.
MATLAB	Reinforcement Learning with Simulink.
NEDUET	Statistical Analysis in Research
NEDUET	Quality Management/ISO 9001:2015 compliance for Higher education